Semester Toets 1	Semestertest 1
Kopiereg voorbehou	Copyright reserved
Elektrisiteit en Elektronika EBN 122	**Electricity and Electronics EBN 122**
Augustus 2018	August 2018

Toetsinligting - Test Information

Maksimum punte: Maximum marks:	45	Volpunte: Full marks:	46
Duur van vraestel: Duration of paper:	90 min + (10 for OCR)	Oop-/Toeboek: Open/Closed book:	Toe/Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (this page included)		16	

Important

1. The examination regulations of the University of Pretoria apply.
2. All questions must be answered on the Optical Character Recognition (OCR) form.
3. Read the instructions on the OCR form carefully.
4. Where applicable, answers must include units.
5. Final answers must be written as real numbers and not as fractions.

Belangrik

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
2. Alle vrae moet op die Optiese Karakter Herkenning (OKH) vorm ingevul word.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
4. Indien van toepassing, moet antwoorde eenhede insluit.
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Lecturers/Dosente:	D. le Roux, J.P. de Villiers, J.P Jacobs

INSTRUCTIONS

Do step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do step 1 in the first 90 minutes of the test. (0-90 min)

Do step 2 within the following 10 minutes. (90-100 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form.

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper will not be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET will not be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.

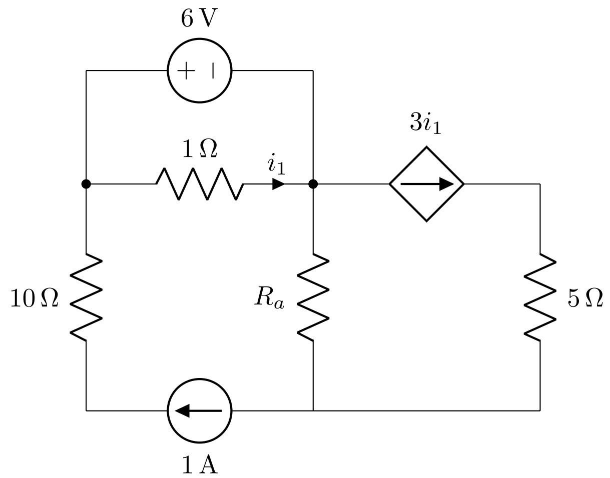
- The Assessment ID is: **2018EBN122S1**
- Use a mechanical pencil with 0.5mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1 / Afdeling 1

Question 1 [2]		Vraag 1 [2]	
1	A student living in an apartment has a weekly electricity usage as follows: • Seven showers (7x2.5kWh), • Twelve meals at home (12x2kW, for 30 minutes), • Two loads of laundry (600W, for 2 hours per load), Calculate the cost of electricity used by the student per month at a cost of R2.50 per kWh. Provide the answer in rands (unit: R), and assume there are exactly four weeks in a month.	**1**	'n Student wat in 'n woonstel bly het 'n weeklikse elektrisiteitsverbruik as volg: • Sewe storte (7x2.5kWh), • Twaalf etes tuis (12x2kW, vir 30 minutes), • Twee ladings wasgoed (600W, vir 2 ure per lading)), Bereken die koste van die maandelikse eleketrisiteitsverbruik van die student teen 'n tarief van R2.50 per kWh. Verskaf die antwoord in rande (eenheid: R) en aanvaar daar is presies vier weke in 'n maand.

Questions 2 to 3 [3]		Vrae 2 tot 3 [3]	
A series circuit consists of voltage source and a 10 Ω resistor. The time varying voltage across the resistor increases linearly from 10 mV to 20 mV between times $t = 0$ and $t = 20$ seconds.		’n Serie-stroombaan bestaan uit ’n spanningsbron en ’n 10 Ω weerstand. Die tydveranderende spanning oor die weerstand neem lineêr toe van 10 mV tot 20 mV tussen $t = 0$ en $t = 20$ sekondes.	
2	Calculate the value of the charge flowing between $t = 10$ and $t = 20$ seconds. [2]	**2**	Bereken die waarde van die lading wat vloei tussen $t = 10$ and $t = 20$ sekondes. [2]
3	Calculate the power associated with the source at $t = 5$ seconds [1].	**3**	Bereken die drywing geassosieer met die bron by $t = 5$ sekondes [1].

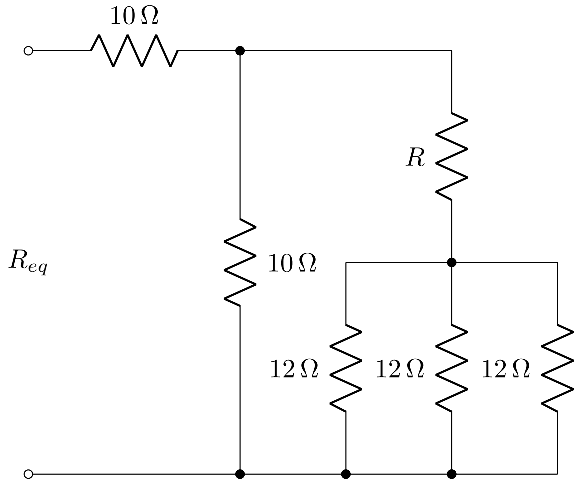
Questions 4 to 5 [4]		**Vrae 4 tot 5 [4]**	
Calculate the power relating to the following elements in the circuit below:		Bereken die drywing geassosieer met die volgende elemente in die onderstaande stroombaan	
4	The independent voltage source. [2]	**4**	Die onafhanklike spanningsbron. [2]
5	R_a if $R_a = 5\ \Omega$. [2]	**5**	R_a indien $R_a = 5\ \Omega$. [2]



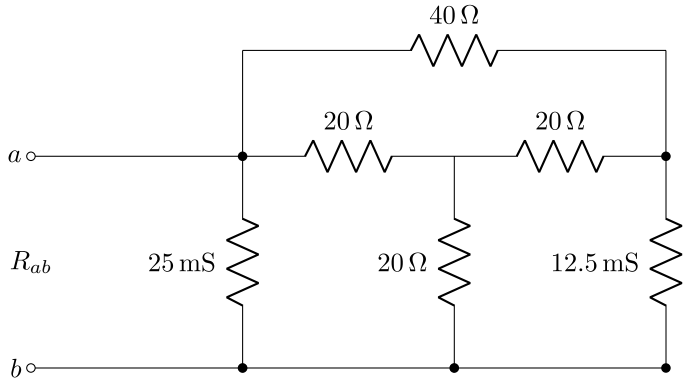
Total Section 1 / Totaal Afdeling 1: [9]

Section 2 / Afdeling 2

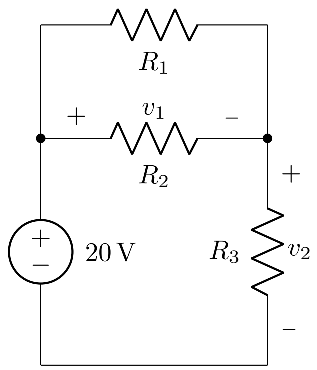
Question 6 [2]		**Vraag 6 [2]**	
6	Determine R if $R_{eq} = 15\ \Omega$.	**6**	Bepaal R indien $R_{eq} = 15\ \Omega$.



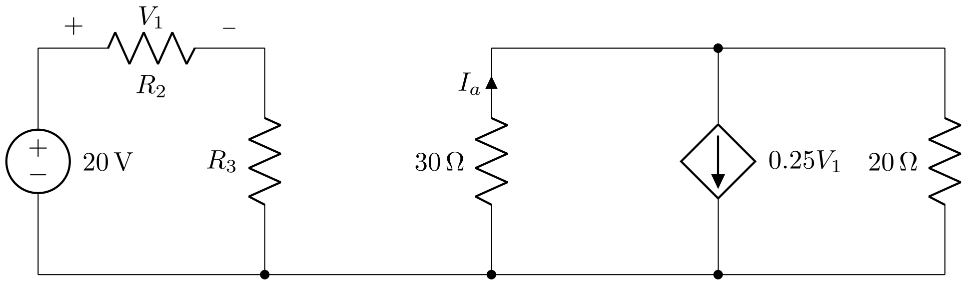
Question 7 [2]		**Vraag 7 [2]**	
7	Determine R_{ab}.	**7**	Bepaal R_{ab}.



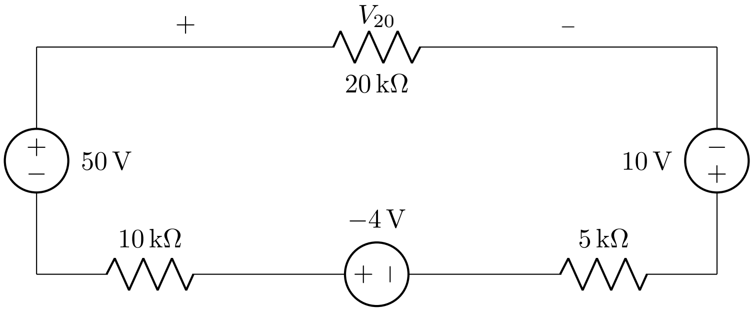
Questions 8 to 10 [5]		**Vrae 8 tot 10 [5]**	
For the circuit below it is given that $R_1\|\|R_2 + R_3 = 40\ \Omega$, $R_1 = 2R_2$ and $v_1/v_2 = 3$. Find R_3, v_1 and v_2		Vir die onderstaande stroombaan word gegee dat $R_1\|\|R_2 + R_3 = 40\ \Omega$, $R_1 = 2R_2$ en $v_1/v_2 = 3$. Bepaal R_3, v_1 en v_2.	
8	R_3 [2]	**8**	R_3 [2]
9	v_1 [2]	**9**	v_1 [2]
10	v_2 [1]	**10**	v_2 [1]



Questions 11 to 12 [4]		Vrae 11 tot 12 [4]	
Determine V_1 and I_a if $R_2 = 3R_3$.		Bepaal V_1 en I_a as $R_2 = 3R_3$.	
11	V_1 [2]	**11**	V_1 [2]
12	I_a [2]	**12**	I_a [2]



Question 13 [2]		Vraag 13 [2]	
13	Determine V_{20} [2]	**13**	Bepaal V_{20} [2]

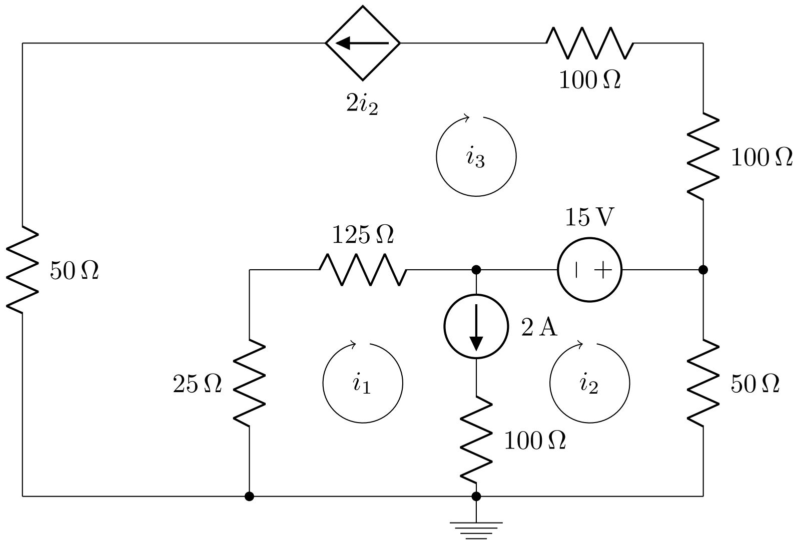


Total Section 2 / Totaal Afdeling 2: [15]

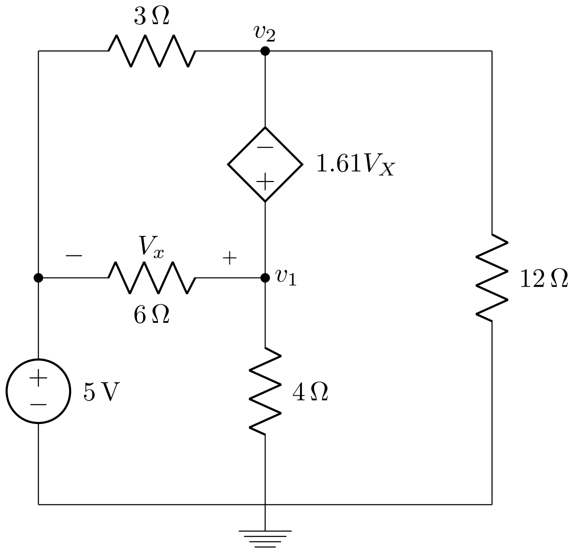
Section 3 / Afdeling 3

Questions 14 to 15 [4]		**Vrae 14 tot 15 [4]**	
14	Determine θ if $v_2 = 6$ V. [2] $$\begin{bmatrix} -4 & -2 \\ 6 & \theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\theta \\ 3\theta \end{bmatrix}$$	**14**	Bepaal θ indien $v_2 = 6$ V. [2] $$\begin{bmatrix} -4 & -2 \\ 6 & \theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -\theta \\ 3\theta \end{bmatrix}$$
15	Determine σ if $i_3 = -1.2$ A. [2] $$\begin{bmatrix} 2 & 1 & 4 \\ 0 & -2 & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2\sigma \end{bmatrix}$$	**15**	Bepaal σ indien $i_3 = -1.2$ A. [2] $$\begin{bmatrix} 2 & 1 & 4 \\ 0 & -2 & 1 \\ 2 & 4 & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2\sigma \end{bmatrix}$$

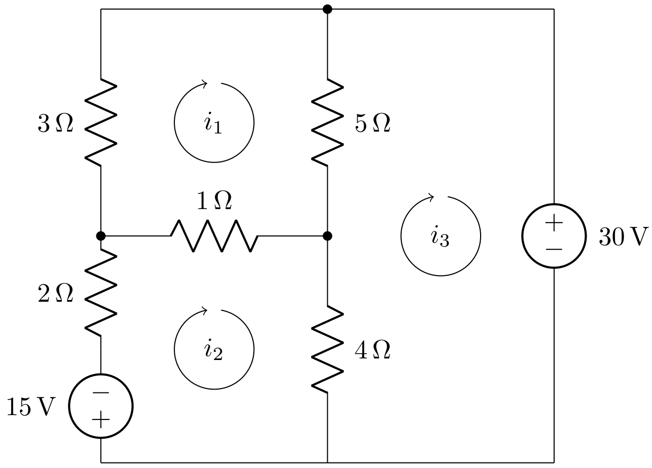
Questions 16 to 19 [4]				Vrae 16 tot 19 [4]			
Use mesh analysis to set up the following two equations of the form: $$ai_1 + bi_2 = 2$$ $$ci_1 + di_2 = 15$$ Calculate the unknown coefficients.				Gebruike lusanalise om die volgende twee vergelykings op te stel met die vorm: $$ai_1 + bi_2 = 2$$ $$ci_1 + di_2 = 15$$ Bepaal die onbekende koëffisiënte.			
16	a [1]	**17**	b [1]	**18**	c [1]	**19**	d [1]



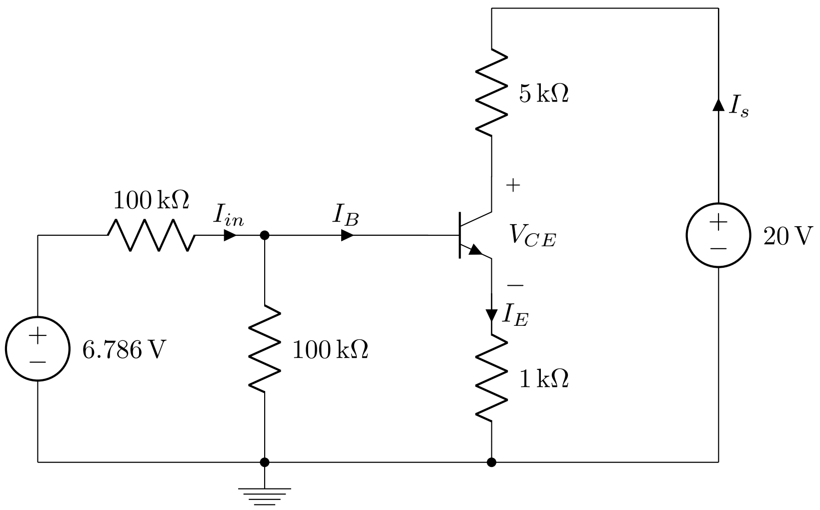
Questions 20 to 23 [4]				**Vrae 20 tot 23 [4]**			
Use nodal analysis to set up the following two equations of the form: $$ev_1 + fv_2 = 60$$ $$gv_1 + hv_2 = -8.05$$ Calculate the unknown coefficients.				Gebruike nodale analise om die volgende twee vergelykings op te stel met die vorm: $$ev_1 + fv_2 = 60$$ $$gv_1 + hv_2 = -8.05$$ Bepaal die onbekende koëffisiënte.			
20	e [1]	**21**	f [1]	**22**	g [1]	**23**	h [1]



Questions 24 to 27 [4]				**Vrae 24 tot 27 [4]**			
Use inspection to compile a matrix equation of the form below $$\begin{bmatrix} 9 & -1 & -5 \\ l & 7 & -4 \\ -5 & k & 9 \end{bmatrix} \mathbf{I} = \begin{bmatrix} p \\ q \\ 30 \end{bmatrix}$$ Then give the values of the following matrix entries.				Gebruike inspeksie om die volgende twee matriksvergelykings op te stel met die vorm: $$\begin{bmatrix} 9 & -1 & -5 \\ l & 7 & -4 \\ -5 & k & 9 \end{bmatrix} \mathbf{I} = \begin{bmatrix} p \\ q \\ 30 \end{bmatrix}$$ Verskaf dan die waardes van die volgende matrikselemente.			
24	k [1]	**25**	l [1]	**26**	p [1]	**27**	q [1]



Questions 28 to 31 [6]		**Vrae 28 tot 31 [6]**	
Consider the case where $I_E = 2.013$ mA and $I_B = 13$ μA in the circuit below. Determine:		Beskou die geval waar $I_E = 2.013$ mA en $I_B = 13$ μA in die stroombaan hieronder. Bepaal:	
28	β [1]	**28**	β [1]
29	I_s [1]	**29**	I_s [1]
30	I_{in} [2]	**30**	I_{in} [2]
31	V_{CE} [2]	**31**	V_{CE} [2]



(Space for rough work / Spasie vir rofwerk)

Total Section 3 / Totaal Afdeling 3: [22]

Grand Total / Groottotaal : [46]

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	8	E	B	N	1	2	2	S	1		

	Answer	Unit
01	Cost =	
02	$Q =$	
03	$p_{5s} =$	
04	$p_{vs} =$	
05	$p_{Ra} =$	
06	$R =$	
07	$R_{ab} =$	
08	$R_3 =$	
09	$v_1 =$	
10	$v_2 =$	
11	$V_1 =$	
12	$I_a =$	
13	$V_{20} =$	
14	$\theta =$	No units
15	$\sigma =$	No units
16	$a =$	No units
17	$b =$	No units
18	$c =$	No units
19	$d =$	No units
20	$e =$	No units
21	$f =$	No units
22	$g =$	No units
23	$h =$	No units
24	$k =$	No units
25	$l =$	No units
26	$p =$	No units
27	$q =$	No units
28	$\beta =$	No units
29	$I_s =$	
30	$I_{in} =$	
31	$V_{CE} =$	